



Cambridge IGCSE™

MATHEMATICS

0580/41

Paper 4 (Extended)

May/June 2022

MARK SCHEME

Maximum Mark: 130

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **13** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles

1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

PUBLISHED**ABBREVIATIONS IN MARK SCHEME**

Abbreviation	Meaning
M	Method marks - for a correct method applied to appropriate numbers.
A	Accuracy marks – depend on M marks. Hence M0 A1 is not possible.
B	Independent of method marks – for a correct final answer, a partially correct answer or a correct intermediate stage.
SC	Marks given in special cases only when indicated in mark scheme.
FT	Work can be followed through after an error.
isw	Ignore subsequent working (after correct answer obtained)
cao	Correct answer only
nfw	Not from wrong working
oe	Or equivalent
soi	Seen or implied
eeo	Each error or omission
dep	Dependent on the previous mark(s)

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Question	Answer											Marks	Partial Marks
1(a)(i)	1	7	7	8	8	9	9					2	B1 for one row correctly ordered or for fully correct unordered stem-and-leaf diagram or for a correct diagram with one error or omission
	2	1	1	1	1	2	3	3	4	5			
1(a)(ii)	21											1	
1(a)(iii)	23											1	
1(a)(iv)	48											2	M1 for $\frac{2}{15}[\times 360]$ or $\frac{360}{15}[\times 2]$
1(b)(i)	120											1	
1(b)(ii)	130											1	
1(b)(iii)	60											1	
1(c)(i)	93.4											4	M1 for mid-values soi M1 for Σfx M1 dep on second M for $\Sigma fx \div 200$
1(c)(ii)	19											2	M1 for $\frac{86}{50}$ or $\frac{114}{60}$

Question	Answer	Marks	Partial Marks
2(a)	42	2	M1 for $12 \div 2$ or better
2(b)(i)	5.72	2	M1 for $\frac{100-12}{100} \times 6.50$ oe or B1 for 0.88 oe
2(b)(ii)	12.5[0]	2	M1 for $\frac{100-12}{100} \times x = 11$ or better oe

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Question	Answer	Marks	Partial Marks
2(c)	4	2	M1 for $\frac{100+2.5}{100} \times [\dots] = \frac{100+6.6}{100}$ oe
2(d)(i)	72.3 or 72.31...	2	M1 for $80 \times \left(\frac{100-2}{100}\right)^5$ oe
2(d)(ii)	4 nfw	3	B2 for answer 9 nfw or M2 for correct trials with values giving either side of 67 or M1 for $80 \times \left(\frac{100-2}{100}\right)^n = 67$ or <i>their</i> (i) $\times \left(\frac{100-2}{100}\right)^k = 67$ or an evaluated trial with $n \geq 6$ or $k \geq 1$

Question	Answer	Marks	Partial Marks
3(a)(i)	6	3	B2 for $4x + 6 = 30$ or better or M1 for $x + x + 7 + 2x - 1 [= 30]$
3(a)(ii)	21	3	M2 for $(555 - \text{their } x \times 15 - \text{their } (x + 7) \times 18) \div \text{their } (2x - 1)$ or M1 for $\text{their } x \times 15$ or $\text{their } (x + 7) \times 18$
3(b)(i)	8	2	M1 for isolating the term in w or correctly removing all fractions e.g. $\frac{3w}{16} = 1 + \frac{1}{2}$ or better or $3w - 16 = 8$
3(b)(ii)	-3	2	M1 for $2^{-y} = 8$ or $2^y = \frac{1}{8}$ or $2^{-y} = \text{their } w$ or better

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Question	Answer	Marks	Partial Marks
3(c)(i)	$[p =] \frac{1}{2}$ oe $[q =] 1$	2	B1 for each If zero scored, SC1 for 2 values satisfying one of the original equations
3(c)(ii)	$[u =] 30$ and 150 $[v =] 0$ and 360	4	B1 for each OR SC1 for $\sin u = \text{their } p$ and $\cos v = \text{their } q$ SC1 if their two different angles for u sum to 180 or if their different two angles for v sum to 360

Question	Answer	Marks	Partial Marks
4(a)(i)	3	1	
4(a)(ii)	7	1	FT <i>their</i> (i) $3 \times \text{their (i)} - 2$
4(b)	$\frac{x+2}{3}$ oe final answer	2	M1 for $y + 2 = 3x$ or $\frac{y}{3} = x - \frac{2}{3}$ or $x = 3y - 2$
4(c)	25	2	M1 for $\frac{1}{x} = 5^{-2}$ oe
4(d)	$\frac{2x^2 - x - 1}{x}$ final answer	2	M1 for $2x - 1 - \frac{1}{x}$
4(e)	2.98×10^{17} or $2.980... \times 10^{17}$	1	
4(f)	625	2	M1 for $x = j(4)$

Question	Answer	Marks	Partial Marks
5(a)(i)(a)	$\frac{(8-2) \times 180}{8 \times 2}$ oe	M2	M1 for $\frac{(8-2) \times 180}{8}$ or $\frac{360}{8}$ or $\frac{(2 \times 8 - 4) \times 90}{8}$
5(a)(i)(b)	174 or 173.8....	4	M3 for $\frac{1}{2} \times 6 \times OM$ oe or $\frac{1}{2} \times (OA)^2 \times \sin 45$ oe or $\frac{1}{2} \times 6 \times OA \times \sin 67.5$ oe where OA and OM are as in the M2 or M2 for $OM = 3 \times \tan 67.5$ oe or for $OA = \left(\frac{3}{\cos 67.5} \right)$ or $\frac{6 \times \sin 67.5}{\sin 45}$ oe or M1 for $\frac{OM}{3} = \tan 67.5$ oe or for $\frac{3}{OA} = \cos 67.5$ oe or for $\frac{\sin 45}{6} = \frac{\sin 67.5}{OA}$ oe
5(a)(ii)	193 or 193.0 to 193.1	3	M2 for $\pi \times \left(\frac{3}{\cos 67.5} \right)^2$ oe or M1 for $\frac{3}{r} = \cos 67.5$ or $\frac{\sin 45}{6} = \frac{\sin 67.5}{r}$

Question	Answer	Marks	Partial Marks
5(b)(i)	1.27 or 1.272 to 1.273	2	M1 for $\left[\frac{1}{2} \times \pi \times 0.45^2 \times 4\right]$ or $\frac{1}{2} \times \pi \times 0.45^2 [\times 4]$
5(b)(ii)	742 or 743	6	M5 for a method leading to the volume of water e.g. $4 \times \left\{ 2 \times \frac{\operatorname{invcos}\left(\frac{0.15}{0.45}\right)}{360} \times \pi \times 0.45^2 \right.$ $\left. - \frac{1}{2} \times 0.45^2 \times \sin\left(2 \operatorname{invcos}\left(\frac{0.15}{0.45}\right)\right) \right\}$ oe OR M2 $[2 \times] \frac{\operatorname{invcos}\left(\frac{0.15}{0.45}\right)}{360} \times \pi \times 0.45^2$ oe or $[2 \times] \frac{90 - \operatorname{invcos}\left(\frac{0.15}{0.45}\right)}{360} \times \pi \times 0.45^2$ oe or M1 for use of $\frac{\theta}{360} \times \pi \times 0.45^2$ oe M2 for $\frac{1}{2} \times 0.45^2 \times \sin\left(2 \operatorname{invcos}\left(\frac{0.15}{0.45}\right)\right)$ oe or $\frac{1}{2} \times 0.15 \times 0.45 \times \sin\left(\operatorname{invcos}\left(\frac{0.15}{0.45}\right)\right) [\times 2]$ oe

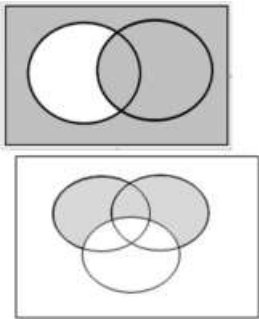
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Question	Answer	Marks	Partial Marks
5(b)(ii)			$\frac{1}{2} \times 0.45^2 \times \sin \theta$ or M1 for use of $\frac{1}{2} \times 0.45^2 \times \sin \theta$ oe $[2 \times] \frac{1}{2} \times 0.15 \times 0.45 \times \sin \beta$ or $\frac{1}{2} \times 0.15 \times 0.45 \times \sin \beta$ oe If 0 scored, SC1 for $\cos^{-1} \left(\frac{0.15}{0.45} \right)$ or $\sin^{-1} \left(\frac{0.15}{0.45} \right)$ or $\sqrt{0.45^2 - 0.15^2}$ soi
6(a)(i)	−3	1	
6(a)(ii)	−1 1.55 to 1.6 4.4 to 4.45	3	B1 for each
6(a)(iii)	−8	1	
6(a)(iv)	Ruled line through origin intersecting curve once	2	B1 for ruled line through origin
6(b)(i)	18	3	B2 for $6x - 12$ or B1 for $6x$ or -12
6(b)(ii)	(2, −5)	2	B1 for each. If 0 scored, M1 for <i>their</i> $6x - 12 = 0$ or states $\frac{dy}{dx} = 0$
6(c)	$[p =] 7$ $[q =] 3$	2	B1 for each

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Question	Answer	Marks	Partial Marks
7(a)	39.6 or 39.57....	4	M2 for $[\cos =] \frac{14^2 + 12^2 - 9^2}{2 \times 14 \times 12}$ or M1 for $9^2 = 14^2 + 12^2 - 2 \times 14 \times 12 \times \cos ACD$ A1 for 0.7708... or 0.771 or $\frac{37}{48}$ oe
7(b)	$\frac{14 \sin 25}{\sin 123}$	M2	M1 for $\frac{\sin 123}{14} = \frac{\sin 25}{BC}$ oe
	7.054...	A1	
7(c)	3.74 or 3.735 to 3.739	3	M2 for $7.05 \times \sin 32$ or M1 for recognition that the line from <i>B</i> is perpendicular to <i>AC</i>
7(d)	11.8 or 11.83 to 11.85	4	M1 for $32 + \text{their}(a)$ soi M2 for $12^2 + 7.05^2 - 2 \times 12 \times 7.05 \times \cos(\text{their}(a) + 32)$ or M1 for $\cos(\text{their}(a) + 32) = \frac{12^2 + 7.05^2 - BD^2}{2 \times 12 \times 7.05}$
7(e)	309.6 or 309.57...	2	FT 270 + <i>their(a)</i> M1 for $270 + \text{their}(a)$ oe

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Question	Answer	Marks	Partial Marks
8(a)(i)	$A \cap B$	1	
8(a)(ii)		2	B1 for each
8(b)(i)	$\frac{9}{11}$	1	
8(b)(ii)	$\frac{36}{121}$ oe	3	M2 for $2 \times \frac{2}{11} \times \frac{9}{11}$ oe or M1 for $\frac{2}{11} \times \frac{9}{11}$ oe If 0 scored SC1 for $\frac{36}{110}$
8(c)(i)	3, 5, 28, 14 correctly placed	2	B1 for 28 in the intersection
8(c)(ii)	$\frac{28}{50}$ oe	1	FT <i>their</i> 28 where <i>their</i> 28 < 50
8(c)(iii)	$\frac{123}{175}$ oe	2	M1 for $\frac{42}{50} \times \frac{41}{49}$
8(c)(iv)	$\frac{63}{88}$ oe	2	FT <i>their</i> 28 M1 for $\frac{\text{their}28}{33} \times \frac{\text{their}28-1}{32}$

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Question	Answer	Marks	Partial Marks
9(a)	$x^2 - x - 30 [= 0]$	B3	M1 for $(2x+1)(x-1) - x^2 = 29$ oe B1 for $(2x+1)(x-1) = 2x^2 - 2x + x - 1$ oe soi
	$(x-6)(x+5)$ oe	M1	or correct factors for <i>their</i> 3 term quadratic equation or for correct substitution into quadratic formula or correctly completing the square for <i>their</i> 3 term quadratic equation
	$x = 6$ cao	B1	
	12 or $2 \times$ <i>their</i> x evaluated or $k = 2x$ stated	B1 FT	
9(b)(i)	$(y+1)^3 - y^3 = 5$ oe	M1	
	$(y+1)^3 = y^3 + 3y^2 + 3y + 1$ soi	B2	B1 for $(y+1)^2 = y^2 + y + y + 1$ oe soi
	Completion to $3y^2 + 3y - 4 = 0$	A1	With no errors or omissions
9(b)(ii)	$\frac{-3 \pm \sqrt{3^2 - 4(3)(-4)}}{2 \times 3}$	B2	or B1 for $\sqrt{3^2 - 4(3)(-4)}$ or for $\frac{-3 + \sqrt{\dots}}{2 \times 3}$ or $\frac{-3 - \sqrt{\dots}}{2 \times 3}$
	0.44	B2	B1 for 0.758 or 0.7583...